Meeting 8 – Part 3

Flash Architecture & Persistent Storage Design

Project: Digital Harmonium Workstation

Version: Final Discussion Draft

Status: Finalized Design

8.3.1 Purpose

The previous meeting established the runtime memory (SRAM) architecture.

This meeting defines the permanent storage architecture.

Unlike SRAM, Flash memory retains information after power is removed.

The purpose of this document is to determine:

- What permanently lives inside the ESP32
- What belongs on the SD card
- How instruments are installed
- How recordings are stored
- How future expansion is supported

The objective is to produce a storage architecture that is deterministic, expandable, easy to maintain, and suitable for a professional embedded musical instrument.

8.3.2 Storage Philosophy

The workstation uses three completely different storage systems.

Each serves a unique purpose.

SRAM

Purpose:

Runtime working memory.

Contains:

- Active voices
- Buffers
- FreeRTOS objects
- LCD state
- Runtime variables

Everything stored here disappears after power loss.

Internal Flash

Purpose:

Permanent installed environment.

Contains:

- Firmware
- Operating software
- Active instrument package
- Configuration
- Calibration
- Persistent settings

Flash behaves like the workstation's internal hard drive.

SD Card

Purpose:

Permanent user library.

Contains:

- Instrument collection
- User recordings
- Presets
- Backups
- Future expansion

The SD card behaves as external long-term storage.

8.3.3 Design Philosophy

The workstation follows a strict installation model.

Only **one active instrument** exists inside Flash at any time.

Every other instrument remains stored on the SD card.

This creates a clean separation between:

Installed software

and

Available library.

8.3.4 Instrument Installation Workflow

When the user selects a new instrument:

```
Instrument Library

↓

SD Card

↓

Instrument Loader

↓

Erase Instrument Partition

↓

Copy Entire VHL

↓

Verification
```

```
↓

Installation Complete

↓

Ready To Play
```

Once installation completes:

The SD card is no longer involved during normal playback.

All playback accesses the internal Flash.

Advantages:

- Lower runtime latency
- Stable playback
- Reduced SD card traffic
- Simpler software architecture
- Predictable timing

8.3.5 Why Entire VHL Files Are Stored

Several possible approaches were considered.

Option 1

Stream directly from SD card.

Advantages:

- Unlimited instrument size

Disadvantages:

- Continuous SD access
- Increased latency
- Greater software complexity
- Possible timing instability

Rejected.

Option 2

Compress VHL before installation.

Advantages:

- Smaller storage footprint

Disadvantages:

- Increased loader complexity
- Compression/decompression logic
- More debugging
- Longer loading times

Rejected.

Option 3 (Final)

Install the complete VHL package into Flash.

Advantages:

- Simplest implementation
- Faster playback
- Cleaner architecture
- No runtime decompression
- Easier debugging
- Future AI agents work directly with original data

Selected.

8.3.6 Flash Memory Layout

The ESP32 provides:

4 MB Internal Flash

This Flash is divided into logical partitions.

Bootloader

Reserved:

64 KB

Purpose:

Initial processor startup.

Loads firmware.

Never modified during normal operation.

Partition Table

Reserved:

4 KB

Purpose:

Defines every Flash partition.

Required by the ESP32 boot process.

NVS (Non-Volatile Storage)

Reserved:

32 KB

Purpose:

Stores persistent system information.

Examples:

- Calibration
- Internal configuration
- ESP32 runtime data

Firmware Partition

Reserved:

1 MB

Purpose:

Contains the complete workstation software.

Includes:

- Operating software
- FreeRTOS
- Audio engine
- UI engine
- Recorder
- Instrument manager
- Storage manager
- Entire application

The firmware remains unchanged during normal operation.

Firmware updates occur only through USB programming.

Instrument Partition

Reserved:

2 MB

Purpose:

Stores the currently installed VHL package.

Contains:

- Complete instrument
- Samples
- Metadata
- Lookup tables
- Regions
- Presets

Only one instrument occupies this partition.

Whenever another instrument is selected:

The previous package is erased.

The new VHL package replaces it.

Settings Partition

Reserved:

64 KB

Purpose:

Stores user preferences.

Examples:

- Last selected instrument
- Transpose
- LCD contrast
- Speaker volume
- Preferred tempo
- Calibration values

These values remain after power loss.

Reserved Partition

Purpose:

Future expansion.

Intentionally left unused.

Provides flexibility for:

- Additional software
- Future effects
- Larger firmware
- New features

Unused Flash is considered good engineering practice.

8.3.7 Flash Partition Summary

Partition	Reserved Size	Purpose
Bootloader	64 KB	Processor startup
Partition Table	4 KB	Flash partition map
NVS	32 KB	System configuration
Firmware	1 MB	Operating software
Instrument Partition	2 MB	Installed VHL package
Settings	64 KB	User configuration
Reserved	Remaining Space	Future expansion

8.3.8 Instrument Header

Every VHL package should begin with a small header.

Example information:

- Instrument Name
- Version
- Creation Date
- Package Size
- Checksum
- CRC
- Author

The loader verifies this header before installation.

If verification fails:

Installation is aborted.

This prevents corrupted instrument packages from being loaded.

8.3.9 SD Card Directory Structure

The SD card serves as the workstation library.

Recommended structure:

```
SD Card

├── Instruments
│
│   ├── Harmonium.vhl
│   ├── Organ.vhl
│   ├── Flute.vhl
│   ├── Future Instruments
│
├── Recordings
│
├── Presets
│
├── Backups
│
├── Firmware Updates
│
└── Documentation
```

Every directory has a dedicated purpose.

This organization simplifies maintenance and future development.

8.3.10 Recording Policy

A major design decision concerns recordings.

Recordings are **never written to Flash.**

Workflow:

```
Performance

↓

Recorder Buffer (RAM)

↓

Background Storage Task
```

```
↓

SD Card
```

Reasons:

- Flash write endurance should be preserved.
- SD cards are designed for frequent file writes.
- Recording does not interfere with firmware or instrument partitions.

8.3.11 Firmware Update Strategy

The workstation does not support OTA (Over-The-Air) updates.

Firmware updates occur only through USB.

Reasons:

- Simpler architecture
- Saves Flash space
- No duplicate firmware partitions
- Lower software complexity

Workflow:

```
PC

↓

USB

↓

Firmware Upload

↓

Flash Programming

↓

Restart
```

8.3.12 Storage Ownership

Each storage device has a clearly defined owner.

Storage	Owner	Responsibility
SRAM	Runtime System	Live execution
Flash	Operating System	Installed environment
SD Card	User	Permanent library

This ownership model prevents overlap between software components.

8.3.13 Expansion Strategy

The architecture supports future instruments without modification.

Example:

Today:

- Harmonium
- Organ

Later:

- Violin
- Santoor
- Shehnai
- Sarangi

Adding new instruments requires only copying additional VHL packages onto the SD card.

No firmware modification is required.

No software redesign is required.

The loader automatically installs whichever instrument the musician selects.

8.3.14 Meeting Summary

This meeting finalized the complete permanent storage architecture of the Digital Harmonium Workstation.

The architecture follows a clean three-level storage hierarchy:

- **SRAM** executes the live runtime.
- **Flash** stores the firmware and the currently installed instrument.
- **SD Card** stores the complete instrument library, recordings, presets, backups, and future content.

By separating runtime memory from permanent storage and archival storage, the workstation achieves low latency, deterministic behavior, simplified software structure, and virtually unlimited future expansion.

This document serves as the permanent reference for all storage-related implementation decisions.

End of Meeting 8 – Part 3: Flash Architecture & Persistent Storage Design